Chapter 6: Use Effective Heuristics for Alpha Sizing

From "Advanced Portfolio Management" by Giuseppe Paleologo

Core Question: How do you convert investment ideas into dollar positions that maximize risk-adjusted returns?

The Sizing Problem

You have expected returns for stocks. Now what?

Complicating factors:

- Stock A: 15% expected return, 30% volatility
- Stock B: 10% expected return, 15% volatility
- Which gets more capital? It's not obvious.

Other considerations:

- How does the existing portfolio affect sizing?
- Your forecasts are imprecise - how does that affect sizing?
- Risk estimates are also imprecise
- What's your actual objective?

Key Concept: Sharpe Ratio

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Sharpe Ratio = Average Return / Volatility
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- Measures risk-adjusted performance
- Real scarce resource is RISK, not capital
- High Sharpe + low volatility → can lever up to hit return target
- Annualized: $SR_{annual} = SR_{daily} \times \sqrt{252}$

Procedure 6.1: Estimate Expected Returns

For each stock:

1. Set investment horizon T (e.g., 3 months)
2. Create price forecast at that horizon
3. Compute total return: $r_{total} = (forecast\_price / current\_price) - 1$

4. Estimate industry return over same horizon
5. Compute idiosyncratic return: $r_{idio} = r_{total} - r_{industry}$
6. Convert to common horizon: $\alpha = r_{idio} / T$

Key insight: Think in IDIOSYNCRATIC returns, not total returns. Factor returns are noise, not signal.

Procedure 6.2: Four Sizing Methods

Given expected returns α and idiosyncratic volatility σ:

Method 1 - Proportional (RECOMMENDED):

$$NMV_i = \kappa \times \alpha_i$$

Method 2 - Risk Parity:

$$NMV_i = \kappa \times \alpha_i / \sigma_i$$

Method 3 - Mean-Variance:

$$NMV_i = \kappa \times \alpha_i / \sigma_i^2$$

Method 4 - Shrinked Mean-Variance:

$$NMV_i = \kappa \times \alpha_i / (p \times \sigma_i^2 + (1-p) \times \sigma^2_{sector})$$

Choose κ so portfolio hits target GMV or volatility.

Critical Finding: Proportional Beats Complex

Empirical results (Table 6.1):

Method	Sharpe Ratio	Loss vs Proportional
Proportional	1.61	—
MV (75% shrink)	1.55	-4%
Risk Parity	1.46	-9%
Mean-Variance	1.07	-34%

Why Mean-Variance fails:

- Estimation error in volatilities
- Low-vol stocks get oversized → mistakes amplified
- Dispersion in volatilities + alpha uncertainty = disaster

Bottom line: Simple proportional rule (NMV $\propto \alpha$) outperforms sophisticated optimization.

Procedure 6.3: Factor-Neutral Sizing

Problem: Raw sizing creates unintended factor exposures.

Solution:

1. Regress α against factor loadings B: $\alpha = B \times x + \varepsilon$
2. Keep residuals ε (factor-neutral alphas)
3. Standardize: $a_i = \varepsilon_i / \Sigma|\varepsilon_i|$
4. Scale to GMV: $NMV_i = a_i \times target\_GMV$

Result: Portfolio with zero factor exposures by construction.

Volatility Targeting (Section 6.6)

Key finding: Target VOLATILITY, not GMV, over time.

Why it works:

- Volatility is persistent (high vol today → high vol tomorrow)
- PnL is NOT predictable
- Reducing exposure when vol spikes → navigate risk without hurting returns

Empirical improvement: +4% to +12% Sharpe improvement depending on sector

Implementation:

- Daily: Scale GMV so predicted volatility matches target
- When vol spikes: Reduce GMV
- When vol drops: Increase GMV

Module Requirements for Sizing Tool

Inputs:

- Expected returns α (from user or model)
- Idiosyncratic volatilities σ (from factor model)
- Factor loadings B (from factor model)
- Target GMV or target volatility

Calculations:

1. Orthogonalize α to factors → factor-neutral alphas
2. Apply proportional sizing rule
3. Scale to target GMV or volatility
4. Calculate resulting factor exposures (should be ~zero)

Outputs:

- Recommended NMV for each stock
- Resulting factor exposures
- Expected portfolio volatility
- Comparison: raw sizing vs factor-neutral sizing

Key Takeaways for Implementation

1. **Use proportional rule** - don't overthink sizing
2. **Orthogonalize first** - remove factor risk at sizing stage
3. **Target volatility over time** - not constant GMV
4. **Simple beats complex** - estimation error dominates theoretical gains